

Proficiency Test 2

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Task 1:

```
for (i in 1:3) {  
  print(mtcars$mpg[i])  
}
```

```
[1] 21  
[1] 21  
[1] 22.8
```

Task 2:

```
count_above <- function(num_vec , cut_off) { #this is defining the function  
  count <- 0 #this is the starting of the count value  
  for (i in 1:length(num_vec)) { #this is where we state the for loop  
    if (num_vec[i] > cut_off) { #this is where we tell that if the num_vec is greater return  
      count <- count + 1 #this is where the it counts the number of ones greater  
    }  
  }  
  return(count) #This returns the count value  
}  
  
count_above(1:31, 15) #this is where we assign num_vec and the cut off values
```

```
[1] 16
```

```
count_above <- function(num_vec , cut_off) {
  count <- 0
  for (i in 1:length(num_vec)) {
    if (num_vec[i] > cut_off) {
      count <- count + 1
    }
  }
  return(count)
}

count_above(iris$Sepal.Length, 6)
```

[1] 61

```
count_above <- function(num_vec, cut_off) {
  count <- 0
  for (i in 1:length(num_vec)) {
    if (num_vec[i] >= cut_off) {
      count <- count + 1
    }
  }
  return(count)
}

count_above(mtcars$disp, 150)
```

[1] 20

Task 3

Masking a function happens when two loaded packages have a function with the same name and the one loaded most recently takes the priority over the earlier one. R warns you about this when you load the package.

Task 4:

```
iris$Sepal.Length[1]
```

```
[1] 5.1
```

This returns the Sepal Length value of the first flower in the data set iris.

```
iris[1,1]
```

```
[1] 5.1
```